

Assignment - 8

1. Create a HTML page with 4 hyperlinks named white, red, blue and pink. The background color of the HTML page will change based on the link which is clicked. Write a JavaScript function to implement it.

```
<!DOCTYPE html>
<html>
<head>
  <title>Change Background Color</title>
  <script>
    function changeColor(color) {
      document.body.style.backgroundColor = color;
    }
  </script>
</head>
<body>
  <h3>Click a color to change background:</h3>
  <a href="#" onclick="changeColor('white')">White</a> |
  <a href="#" onclick="changeColor('red')">Red</a> |
  <a href="#" onclick="changeColor('blue')">Blue</a> |
  <a href="#" onclick="changeColor('pink')">Pink</a>
</body>
</html>
```

2. Write a JavaScript program which will
 - a. give an alert message of today's date.
 - b. prompt the user to give their name and gives an alert message 'Hello' and username.
 - c. prompt the user to give 2 numbers and alerts the sum of those 2 numbers.

```
<!DOCTYPE html>
<html>
<body>
  <script>
    // a. Alert today's date
    alert("Today's Date: " + new Date().toString());

    // b. Prompt for name
    let name = prompt("Enter your name:");
    alert("Hello " + name + "!");

    // c. Prompt for two numbers and show sum
    let num1 = parseFloat(prompt("Enter first number:"));
    let num2 = parseFloat(prompt("Enter second number:"));
    alert("Sum = " + (num1 + num2));
  </script>
</body>
</html>
```

3. Write a JavaScript program that prompts the user to enter a number and display whether the number given is odd or even.

```
<!DOCTYPE html>
<html>
<body>
  <script>
    let num = prompt("Enter a number:");
    if (num % 2 == 0) {
      alert(num + " is Even");
    } else {
      alert(num + " is Odd");
    }
  </script>
</body>
</html>
```

4. Create program to display current time in a textbox (HH:MM:SS) such that the value in the time textbox is dynamic (Time should get updated every second) and not static. [Hint: Use setTimeout function]

```
<!DOCTYPE html>
<html>
<body>
  <h3>Dynamic Clock</h3>
  <input type="text" id="clock" readonly>

  <script>
    function updateTime() {
      let now = new Date();
      let time = now.toLocaleTimeString();
      document.getElementById("clock").value = time;
      setTimeout(updateTime, 1000);
    }
    updateTime();
  </script>
</body>
</html>
```

5. Write validation functions and modify the onSubmit event Handler in the form code to validate the following form fields:
1. Member number
 - Must be entered
 - Must be a number
 2. Password
 - Must be entered
 - Must be longer than 4 characters
- If an error occurs, use an alert box to display an error message for the first field in error and place the cursor in the first field in error.

```
<!DOCTYPE html>
<html>
<body>
  <form onsubmit="return validateForm()">
    <h3>Member Login</h3>
    Member Number: <input type="text" id="member"><br><br>
    Password: <input type="password" id="password"><br><br>
    <input type="submit" value="Submit">
  </form>

  <script>
    function validateForm() {
      let member = document.getElementById("member");
      let password = document.getElementById("password");

      if (member.value.trim() === "") {
        alert("Member number must be entered");
        member.focus();
        return false;
      }

      if (isNaN(member.value)) {
        alert("Member number must be a number");
        member.focus();
        return false;
      }

      if (password.value.trim() === "") {
        alert("Password must be entered");
        password.focus();
        return false;
      }

      if (password.value.length <= 4) {
        alert("Password must be longer than 4 characters");
        password.focus();
        return false;
      }
    }
  </script>
</body>
</html>
```

```

        alert("Form submitted successfully!");
        return true;
    }
</script>
</body>
</html>

```

6. Write validation functions and modify the onSubmit event Handler in the form code to validate the following form fields:

Phone Number

- Must be entered
- Must be in below format

XXX-XXX-XXXX

XXX.XXX.XXXX

```

<!DOCTYPE html>
<html>
<body>
    <form onsubmit="return validatePhone()">
        <h3>Phone Number Validation</h3>
        Phone Number: <input type="text" id="phone" placeholder="XXX-XXX-XXXX or XXX.XXX.XXXX"><br><br>
        <input type="submit" value="Submit">
    </form>
    <script>
        function validatePhone() {
            let phone = document.getElementById("phone");
            let pattern = /^d{3}[-.]d{3}[-.]d{4}$/;

            if (phone.value.trim() === "") {
                alert("Phone number must be entered");
                phone.focus();
                return false;
            }

            if (!pattern.test(phone.value)) {
                alert("Phone number must be in format XXX-XXX-XXXX or XXX.XXX.XXXX");
                phone.focus();
                return false;
            }

            alert("Phone number is valid!");
            return true;
        }
    </script>
</body>
</html>

```